INCLUSIONS OF ALL THE STANDARD COVERING CLASSES IN THE V COVERING CLASS

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file topologies.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): the official pool candidate recorded statement [3845, 3850] and proof(s) [[3852, 3865]]; this spec covers the complete environment instead (statement 3845-3850, proof span(s) [(3852, 3865)]), since in this chapter the author's proof environment(s) follow the statement environment and some pool entries under-counted them.

Source reference. Original lemma 022E: sources/stacks-chow/tags.txt, source lines 2703–2703. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0ETC: sources/stacks-chow/tags.txt, source lines 18512–18512. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 022C: sources/stacks-chow/tags.txt, source lines 2701–2701. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0ETD: sources/stacks-chow/tags.txt, source lines 18513–18513. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Any fpqc covering is a V covering. A fortiori, any fppf, syntomic, smooth, étale or Zariski covering is a V covering. Also, a ph covering is a V covering.*

Proof. An fpqc covering can affine locally be refined by a standard fpqc covering, see Lemmas 022E. A standard fpqc covering is a standard V covering, see Lemma 0ETC. Hence the first statement follows from our definition of V covers in terms of standard V coverings. The conclusion for fppf, syntomic, smooth, étale or Zariski coverings follows as these are fpqc coverings, see Lemma 022C.

The statement on ph coverings follows from Lemma 0ETD in the same manner. □

Date: stacks-project:topologies.tex:local-snapshot-20261012; GNU FDL 1.2 or later.